

INQUIRY-BASED LEARNING

How to Use the 5E Model in Your Science Classroom

An inquiry-focused method gives students a way to connect scientific ideas to their experiences and apply their learning.

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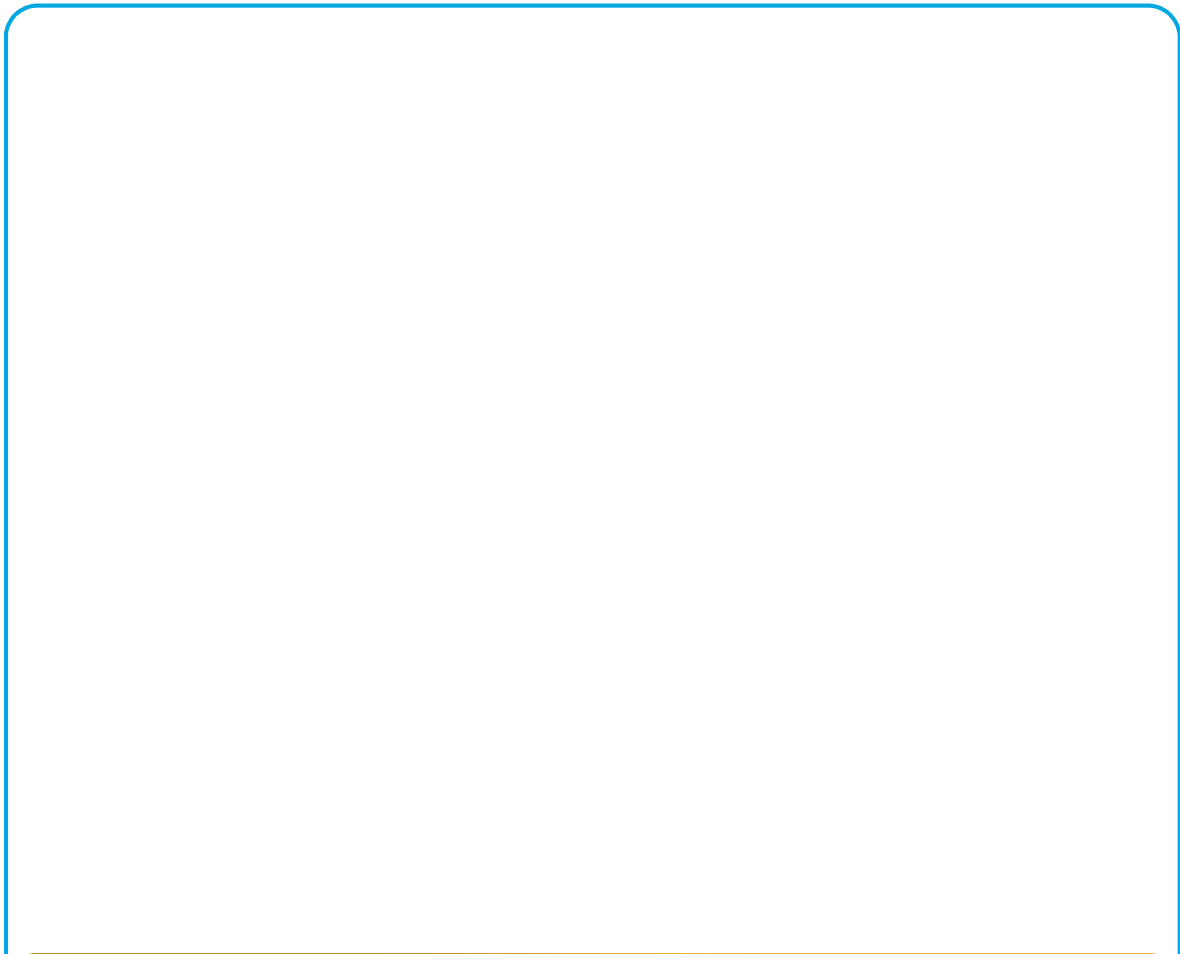
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Try to recall the last science class you taught. It probably had many components over the course of 50 minutes, such as a video, a mini-lecture, an assessment, discussion, or a demonstration. What were all the components? Did your students discuss a question in small groups? Was it a whole-class discussion? Did you introduce a new concept? In what order did you introduce these components? How did you decide what came first? What is the optimal order of activities in an inquiry-based science classroom?

ENGAGE STUDENTS' CURIOSITY

Teaching science using an inquiry approach requires instructional planning that encourages students to engage their curiosity to ask questions, explore solutions to socio-scientific issues, use evidence-based explanations to justify their reasoning, elaborate on possible effects, evaluate their findings, and predict potential outcomes based on different variables. In inquiry science, students are cognitively challenged as they engage in authentic problems while learning content, practicing reasoning skills, and communicating their ideas.

One approach to inquiry science is the 5E instructional model (Engage, Explore, Explain, Elaborate, Evaluate). The 5E model is a planning tool for inquiry teaching that provides a structure for students to connect science ideas with their experiences and apply their learning to new contexts. The 5E model comprises five phases that help teachers build a sequence of coherent and engaging learning experiences for students.





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HOW THE 5E MODEL WORKS

1. Engage. The teacher uses short activities to promote curiosity. The activity must connect prior knowledge to new learning experiences in order to expose any misconceptions and prepare students for new learning.

Novel questions, discrepant events, demonstrations, or a powerful visual are ideal ways to engage students and ascertain their prior knowledge or any misconceptions that might interfere with constructing new knowledge. For new concepts to become meaningful, students use their prior knowledge to connect their past and present learning experiences. The engagement phase doesn't have to be part of the class time. It can be structured as a homework assignment where students can read an article related to the new topic to be introduced, explore a website, watch a video, or answer a question related to their prior knowledge.

For example: *Why are acidic drinks stored in a cold place?*

2. Explore. A lab investigation or hands-on activities are usually introduced in this phase as students attempt to investigate a problem. Conflicting ideas, questions, and confusion are common and help students identify what they need to know before new terms or concepts are introduced in the Explain phase.

Encourage students to think of the following questions:

- What is the problem I am trying to solve?
- What do I need to find out?
- What do I know already?

Students are provided with two identical soda cans, a bottle opener, hot water, and an ice bath. Students perform the activity in pairs or groups, write down their observations, and discuss their results in the group.

For example: *Which soda, the warm or the cold one, had more dissolved carbon dioxide? List all the ways that you know.*

3. Explain. With the teacher's guidance, students explain the concepts they explored in the previous phase and demonstrate their understanding of the new terms that were introduced. Depending on the topic and the grade level, teacher-led instruction might be necessary to address any confusion and questions that came up in the Explore phase. Questions can make learning more meaningful, interactive, and participatory.

Students provide a tentative explanation as they present their results to the whole class. The teacher may pose additional questions to support the discussion. It is recommended that the teacher introduce a formative assessment in this phase to identify if the students are ready to start the Elaborate phase and determine if additional instruction or assistance is needed.

For example: *What is the best way to store an opened bottle of soda so that it doesn't go flat quickly?*

4. Elaborate. Students apply their knowledge to new experiences and extend their conceptual understanding as they solve a problem in a new context before evaluation in the last phase of the 5E model. Elaboration activities can take place during classroom time, or they can be a homework assignment.

For example: *Decompression sickness (DCS) occurs when divers swim to the surface too quickly (rapid ascent). What causes DCS to occur?*

5. Evaluate. Students evaluate their learning and demonstrate their understanding and mastery of key concepts. Evaluation doesn't have to be limited to a quiz or test. It can be a product such as a presentation, a poster, a pamphlet, a journal article, or a final paper.

For example: *In your opinion, why do fish wash ashore on hot summer days?*

Many power plants condense steam by pumping cool river or lake water around the steam pipes. The steam cools and condenses as its heat is transferred to the water, which is then returned to the river or lake. What impact does this warm water have on the fish in the lake or river?

You may already be using these elements without being familiar with the 5E model's formal structure. Consider the phases discussed above and reflect on the activities you planned for the last science lesson you taught. Which component would you classify as an Engage activity? Explore? Explain? How well does the order you have planned reflect the 5E model?

The goal isn't to plan every science lesson according to the 5E model—it's to consider the order and sequence of activities to align with the model to maximize student learning.

Here are some final suggestions:

Start small. You might design a lesson with only two components of the 5E model. If your lesson already has a hands-on activity, you may want to start with Engage to get the students thinking about the hands-on activity they're going to complete in the Explore phase. A simple 3- to 5-minute activity such as a current events story, a video, an advertisement, a problem scenario, or a challenge statement based on a common misconception can engage students.

Explore before Explain. Lab investigations or hands-on activities in the Explore phase, no matter how simple, may be time-consuming. Yet, you can allow at least some exploration before explanation to prepare students to receive new information. They might attempt to solve a problem, make a prediction about an experiment or demonstration, or answer a complex question. Consider starting instruction (Explain) in the middle of the class session, as opposed to the beginning, after students have had time to do some exploration.

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